5 Architecture

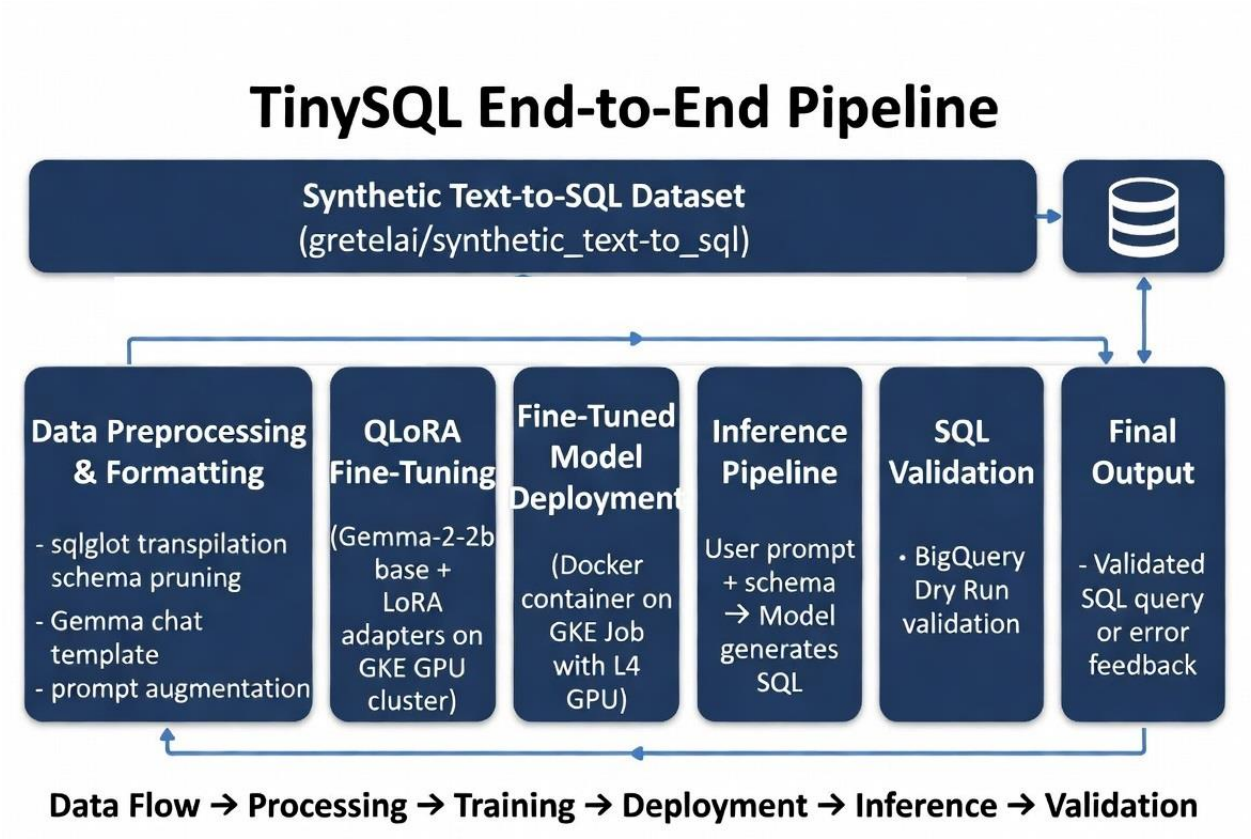
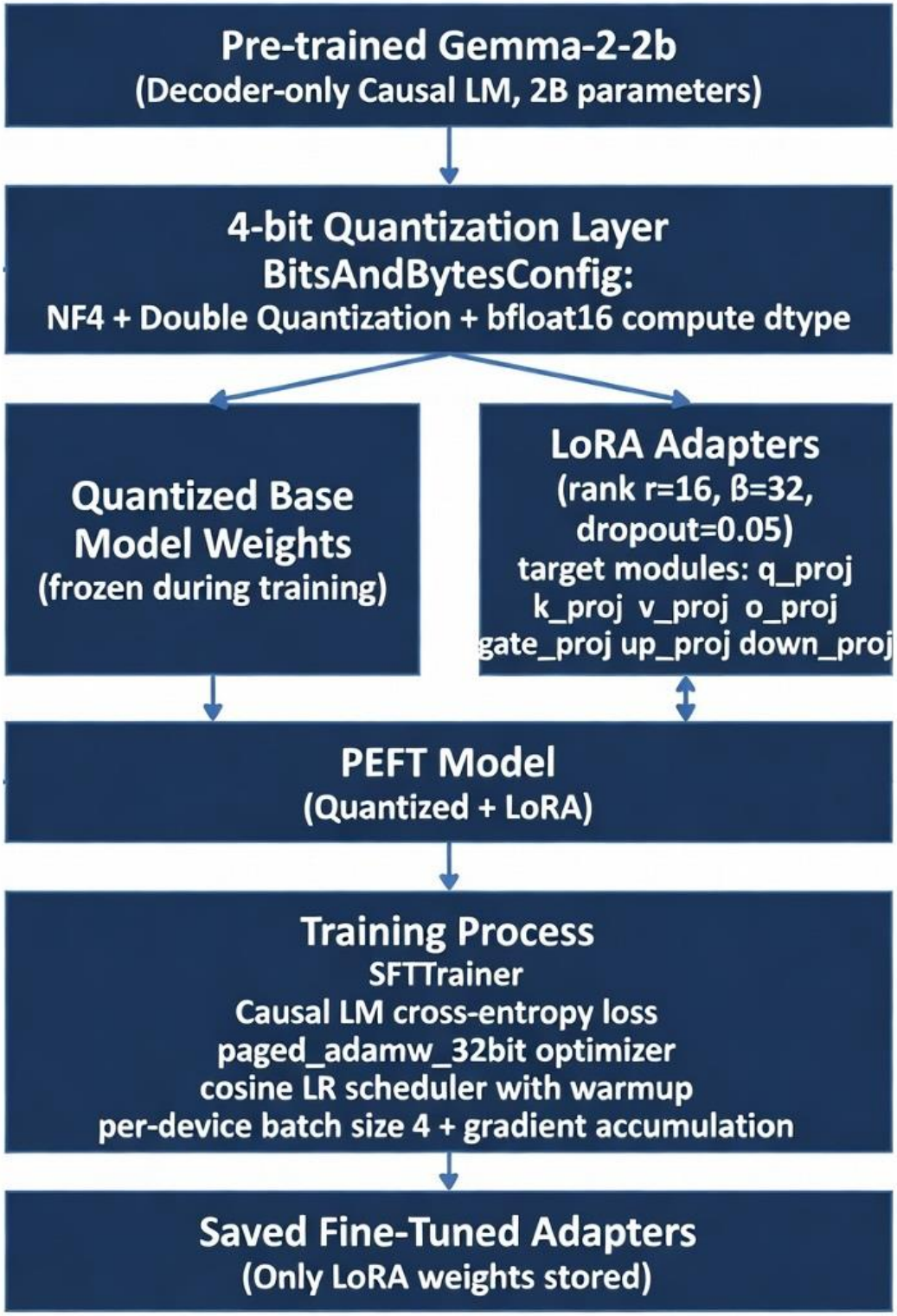
The final model for this text-to-SQL generation system is a decoder-only causal language model based on the Gemma-2-2b architecture. The pre-trained weights were loaded in 4-bit quantized form using BitsAndBytes to enable training on a single NVIDIA L4 GPU while preserving numerical stability through NF4 quantization, double quantization, and bfloat16 compute dtype. Parameter-efficient fine-tuning was performed with Low-Rank Adaptation (LoRA) configured at rank 16, scaling factor $\alpha = 32$, dropout 0.05, and no bias terms. Adaptation targeted the core transformer projection modules (q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, and down_proj) because these layers govern attention and feed-forward behavior and therefore provide the highest impact on SQL syntax generation with minimal additional trainable parameters.

The machine learning pipeline started with the gretelai/synthetic_text_to_sql dataset. Records were preprocessed through a custom formatting function that (1) transpiled original SQL to GoogleSQL-compatible syntax and stripped constraints using sqlglot, (2) prefixed all table references with the target BigQuery dataset identifier, and (3) wrapped each example in the exact Gemma chat template used at inference time, including the full schema, natural-language question, and gold SQL as the model continuation. The dataset was split once via train_test_split (test_size = 1 000, seed = 42); the training partition was further subsampled to 20 000 examples and the validation partition retained as-is. No additional feature engineering beyond schema pruning (occasional distractor tables) and prompt-style augmentation was required.

Training was executed with the SFTTrainer from the TRL library. The procedure used a per-device batch size of 4, gradient accumulation steps of 2 (effective batch size 8), maximum sequence length of 1 024 tokens, and the paged_adamw_32bit optimizer. A cosine learning-rate scheduler with 0.03 warmup ratio and initial learning rate 2×10^{-4} was applied for one epoch, with evaluation every 200 steps. The loss function was standard causal-language-model cross-entropy, optimized for next-token prediction of the target SQL continuation.

During model optimization, LoRA rank, alpha, dropout, and training hyperparameters (batch size, gradient accumulation, warmup ratio, and scheduler type) were adjusted according to the values defined in config.yaml and refined in the final training script. The final model architecture and pipeline therefore consist of the 4-bit quantized Gemma-2-2b base, the LoRA configuration with rank 16 and $\alpha = 32$ targeting the seven projection modules listed above, and the SFTConfig parameters described. All

training and inference were containerized and executed on a GKE cluster with GPU node pools, but these infrastructure details are orthogonal to the model and training procedure itself.



QLoRA Architectural Integration (Gemma-2-2b)